

Ingénierie des circuits intégrés

MASTER IESE

2025-2026

Plan

Part Forth:

- Current Mirrors
- Basic cells of Amplifiers
- Small signal and dynamic region schematic representation
- Input/output resistance of basic cells amplifiers
- Gain and amplification

Part five:

- Digital circuit engineering

IC Biasing—Current Sources, Current Mirrors

MOS Current-Steering Circuits

- The reference current I_{REF} determined by the transistor Q_1 and Resistance R
- Transistors Q_1 , Q_2 , and Q_3 form a two-output current mirror,
- To ensure operation in the saturation region, the voltages at the drains of Q_2 and Q_3 are constrained as follows:

$$V_{D2}, V_{D3} \geq -V_{SS} + V_{GS1} - V_{tn}$$

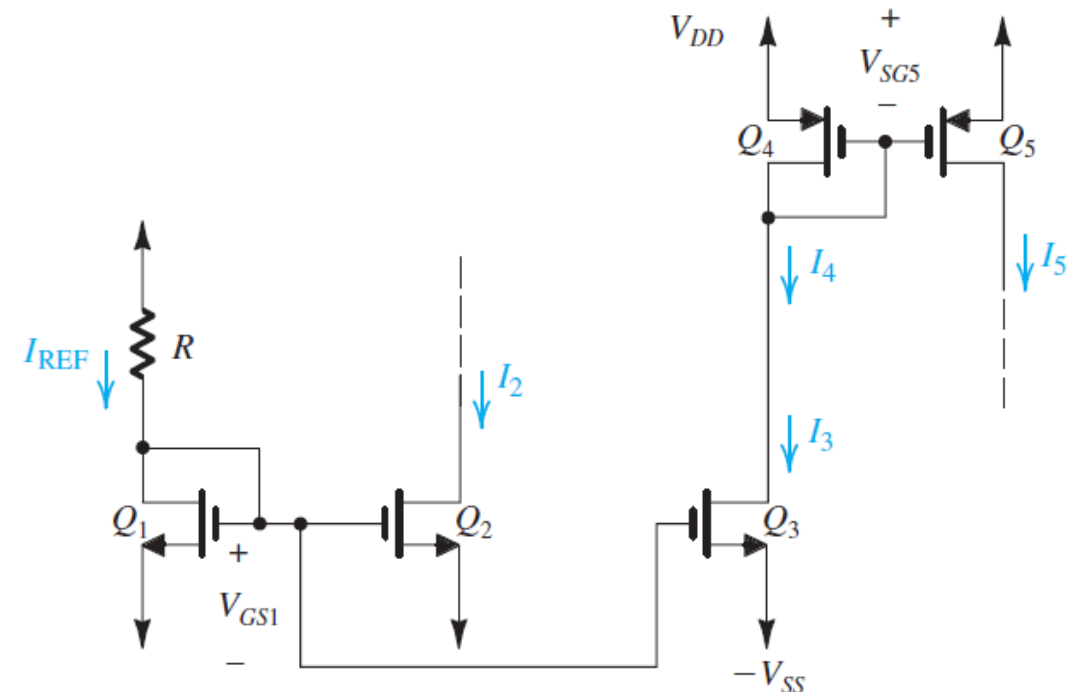
with

$$I_2 = I_{REF} \frac{(W/L)_2}{(W/L)_1} \quad \text{and} \quad I_5 = I_4 \frac{(W/L)_5}{(W/L)_4}$$

$$I_3 = I_{REF} \frac{(W/L)_3}{(W/L)_1}$$

For keeping Q_5 in saturation, its drain voltage should be

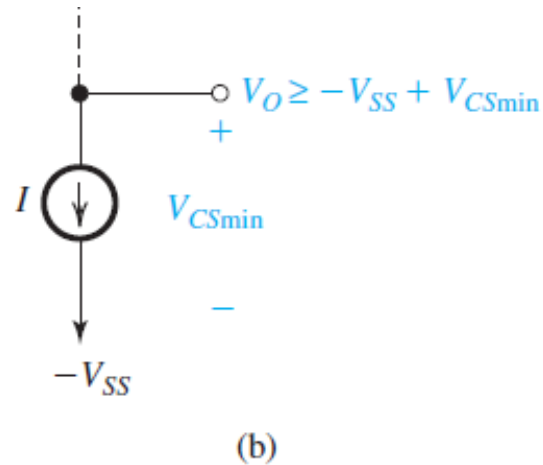
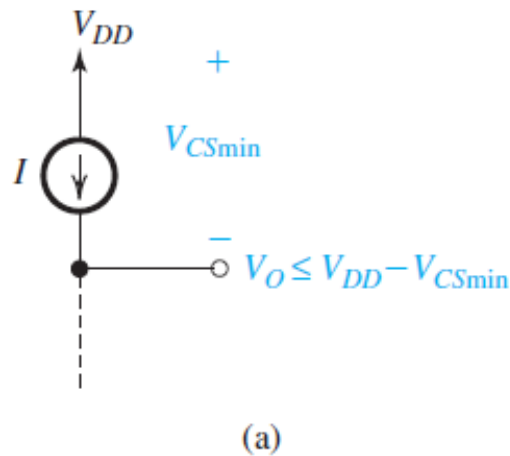
$$V_{D5} \leq V_{DD} - |V_{OV5}|$$



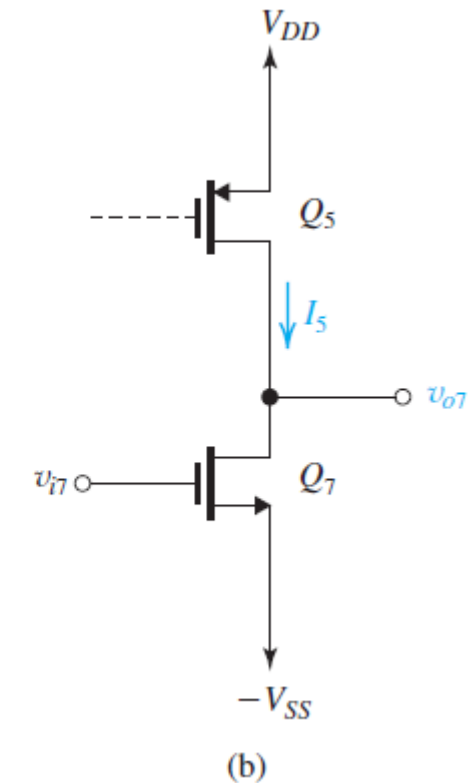
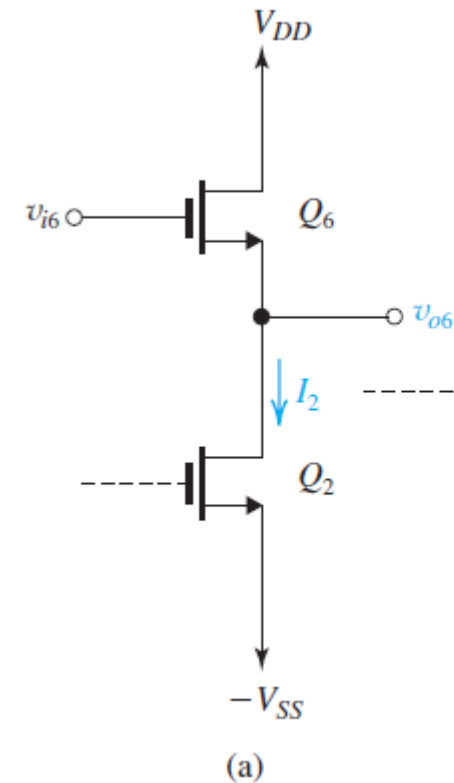
IC Biasing—Current Sources, Current Mirrors

Application of MOS Current-Steering Circuits

- Q2 pulls its current I_2 from a circuit,
- Q2 should more properly be called a **current sink**
- Q5 pushes its current I_5 into a circuit
- Q5 is appropriately called a **current source**,



(a) A current source; and (b) a current sink.



Application of current-steering circuit, a) source follower stage, b) common source stage

IC Biasing—Current Sources, Current Mirrors

Small-Signal Operation of Current Mirrors

- Biasing, current mirrors as current amplifiers.
- Used to derive the small-signal parameters of the current mirror, that is, the input resistor R_{in} , the current gain A_{is} , and the output resistor R_o .

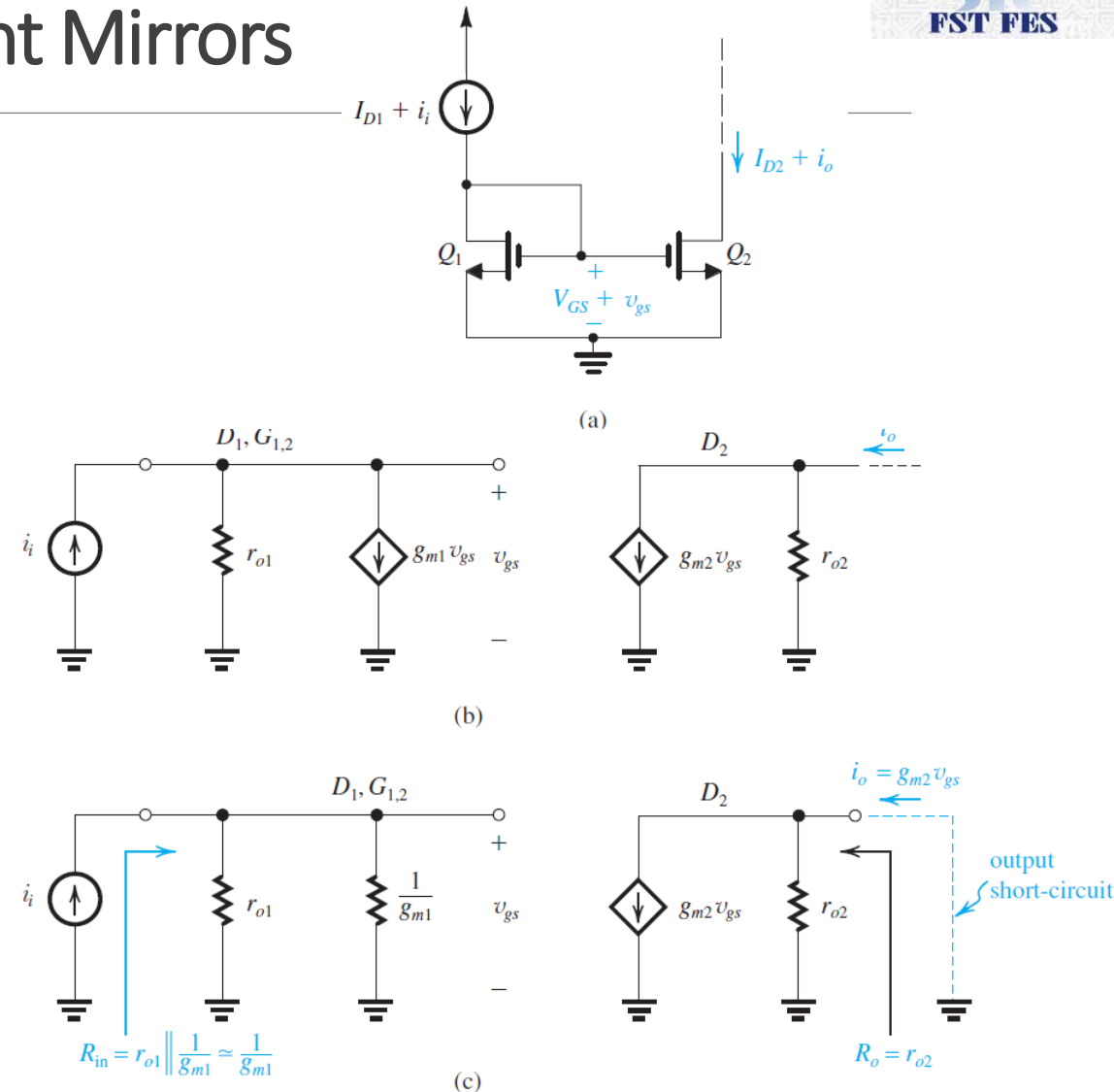
$$R_{in} = r_{o1} \parallel \frac{1}{g_{m1}} \simeq \frac{1}{g_{m1}}$$

$$R_o = r_{o2}$$

$$A_{is} \equiv \left. \frac{i_o}{i_i} \right|_{v_{d2}=0} = \frac{g_{m2} v_{gs}}{i_i} \simeq \frac{g_{m2} i_i / g_{m1}}{i_i} \quad A_{is} = \frac{g_{m2}}{g_{m1}}$$

With $g_{m1,2} = \mu_n C_{ox} (W/L)_{1,2} V_{OV1}$

$$A_{is} = \frac{(W/L)_2}{(W/L)_1}$$



a) Current mirror in small signal, b) small signal operation, b) small signal parameters

Basic cell/Amplifier

The Basic cell Amplifiers

Amplification is an essential function in most analog circuits !

- Why do we amplify a signal ?
- The signal is too small to drive a load
- To overcome the noise of a subsequent stage
- Amplification plays a critical role in feedback systems

In this document we will focus on :

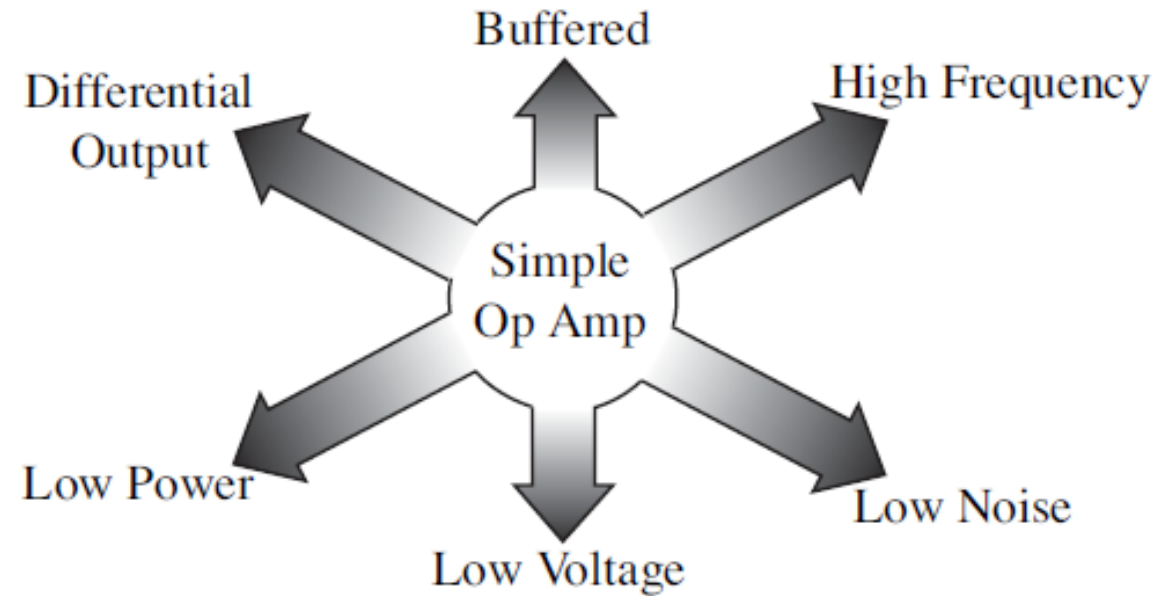
- Low frequency behavior of single stage CMOS amplifiers:
- Common Source, Common Gate, Source Follower,
- Large and small signal analysis.

Amplifier in CMOS TEchnology

Application of MOS Transistor

Goal

To illustrate the degrees of freedom and choices of different circuit architectures that can enhance the performance of a given op amp.



The Basic cell Amplifiers

Approximation of a nonlinear system

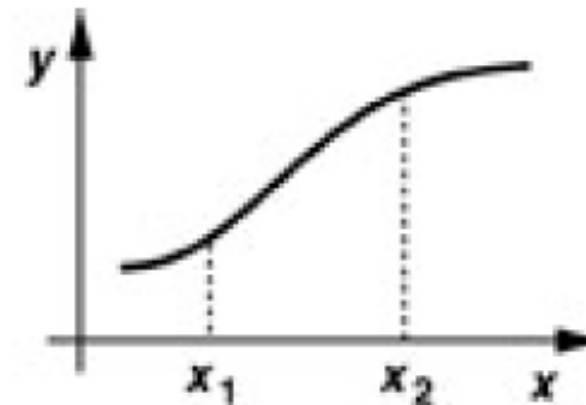
Input-Output Characteristic of a nonlinear system

$$y(t) \approx \alpha_0 + \alpha_1 x(t) + \alpha_2 x^2(t) + \dots + \alpha_n x^n(t) \quad x_1 \leq x \leq x_2$$

In a sufficiently narrow range:

$$y(t) \approx \alpha_0 + \alpha_1 x(t)$$

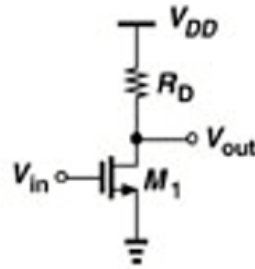
where α_0 can be considered
the operating (bias) point and
 α_1 the small signal gain



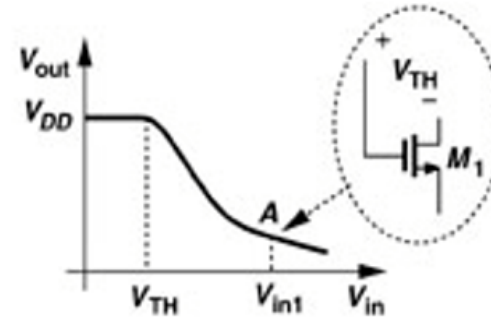
Basic cell/ The CS Amplifier

Common Source Stage with Resistive Load

$$V_{in} = V_{GS}$$



(a)



(b)

$$V_{out} = V_{DD} - R_D I_D$$

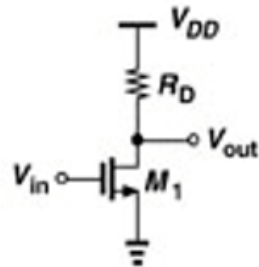
M1 in the saturation region:
$$V_{out} = V_{DD} - R_D \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_{in} - V_{TH})^2$$

M1 in limit of saturation:
$$V_{in1} - V_{TH} = V_{DD} - R_D \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_{in1} - V_{TH})^2$$

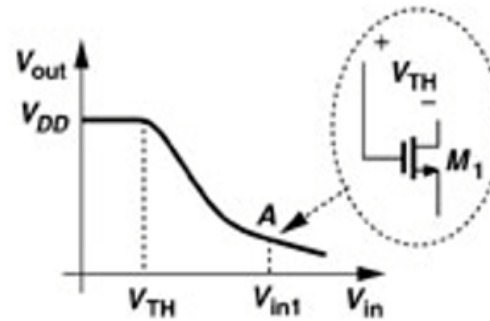
M1 in the linear region:
$$V_{out} = V_{DD} - R_D \mu_n C_{ox} \frac{W}{L} \left[(V_{in} - V_{TH}) V_{out} - \frac{V_{out}^2}{2} \right]$$

Basic cell/ The CS Amplifier

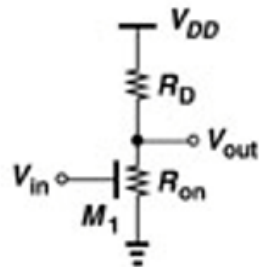
Common Source Stage with Resistive Load



(a)



(b)



(c)

M1 in deep linear region:

$$V_{out} = V_{DD} \frac{R_{on}}{R_{on} + R_D} = \frac{V_{DD}}{1 + \mu_n C_{ox} \frac{W}{L} R_D (V_{in} - V_{TH})}$$

Basic cell/ The CS Amplifier

Common Source Stage with Resistive Load

M1 in the saturation region:

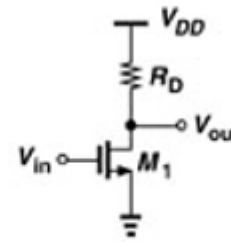
$$V_{out} = V_{DD} - R_D \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_{in} - V_{TH})^2$$

Small signal gain:

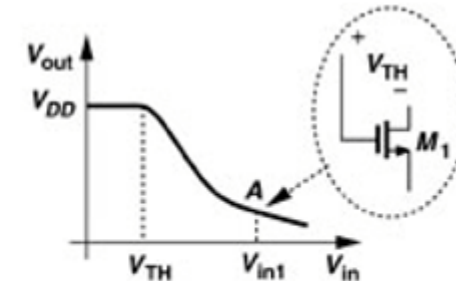
$$A_v = \frac{\partial V_{out}}{\partial V_{in}} = -R_D \mu_n C_{ox} \frac{W}{L} (V_{in} - V_{TH})$$

$$= -g_m R_D$$

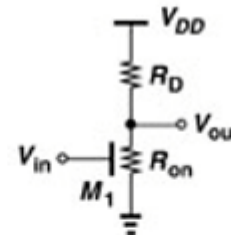
Same relation can be derived from the small signal equivalent circuit



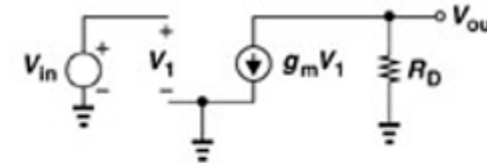
(a)



(b)



(c)

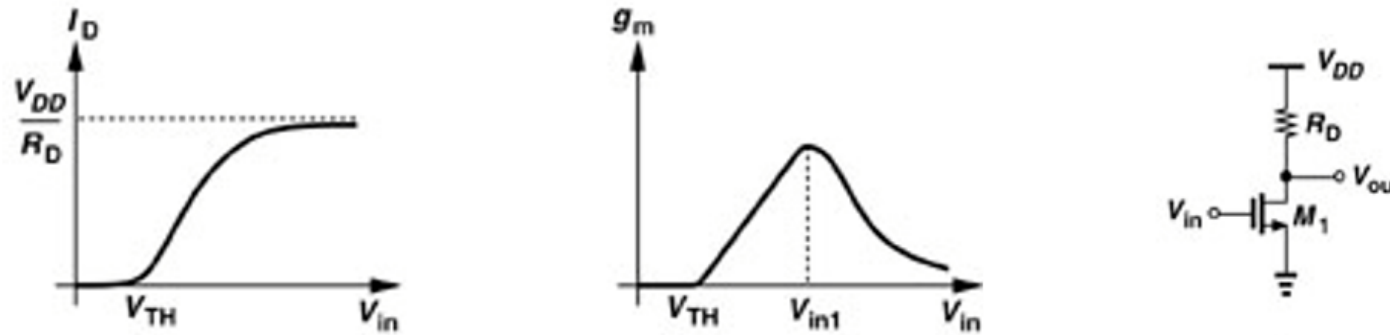


Small signal model for the saturation region

Basic cell/ The CS Amplifier

Common Source Stage with Resistive Load

Sketch I_D and g_m of M1 as a function of the V_{in} :



• M1 in the saturation region:

$$I_D = \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_{in} - V_{TH})^2$$

$$g_m = \frac{\partial I_D}{\partial V_{GS}} = \mu_n C_{ox} \frac{W}{L} (V_{in} - V_{TH})$$

• M1 in the linear region:

$$I_D = \mu_n C_{ox} \frac{W}{L} \left[(V_{in1} - V_{TH}) V_{out} - \frac{V_{out}^2}{2} \right]$$

$$g_m = \frac{\partial I_D}{\partial V_{GS}} = \mu_n C_{ox} \frac{W}{L} V_{out}$$

$$V_{out} = \frac{V_{DD}}{1 + \mu_n C_{ox} \frac{W}{L} R_D (V_{in} - V_{TH})}$$

CS amplifier gain in saturation region

$$A_v = -g_m R_D$$

How to increase A_v ?

Trade-offs:

- Increase W/L → Greater device capacitances.
- Increase V_{RD} → Limits V_{out} swing.
- Reduce I_D → Greater Time Constant.

Basic cell/ The CS Amplifier

Taking Channel Length Modulation into account

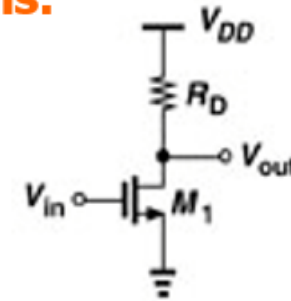
Calculating A_v starting from the Large Signal Equations:

$$V_{out} = V_{DD} - R_D \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_{in} - V_{TH})^2 (1 + \lambda V_{out})$$

$$A_v = \frac{\partial V_{out}}{\partial V_{in}} = -R_D \mu_n C_{ox} \frac{W}{L} (V_{in} - V_{TH}) (1 + \lambda V_{out}) - R_D \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_{in} - V_{TH})^2 \lambda \frac{\partial V_{out}}{\partial V_{in}}$$

$$A_v = -R_D g_m - R_D I_D \lambda A_v$$

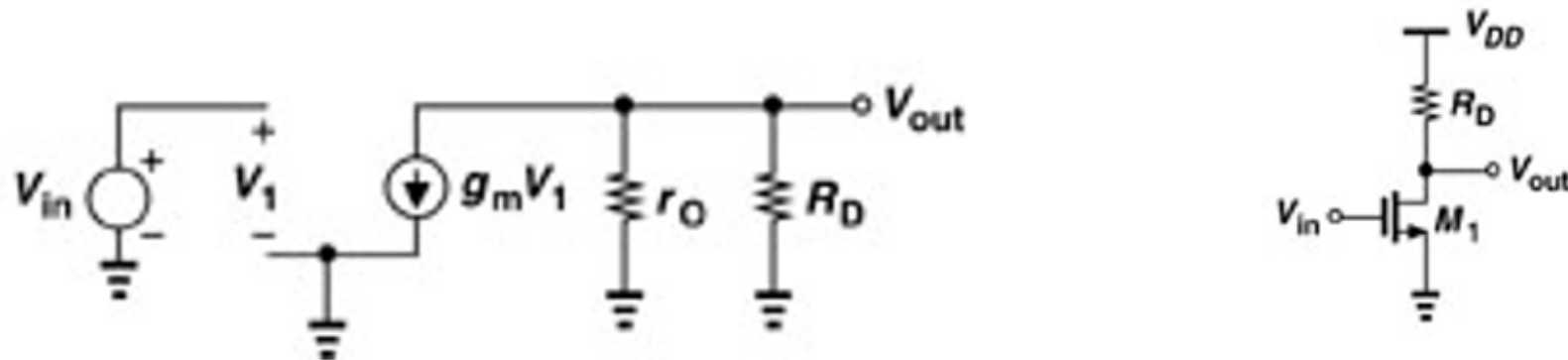
$$A_v = -\frac{R_D g_m}{1 + R_D \lambda I_D} \xrightarrow{\lambda I_D = 1/r_o} \boxed{A_v = -g_m \frac{r_o R_D}{r_o + R_D}}$$



Basic cell/ The CS Amplifier

Taking Channel Length Modulation into account

Calculating A_v starting from the Small Signal model:



$$\left. \begin{aligned} g_m V_1 (r_o // R_D) &= -V_{out} \\ V_1 &= V_{in} \end{aligned} \right\} \boxed{A_v = \frac{V_{out}}{V_{in}} = -g_m (r_o // R_D)}$$

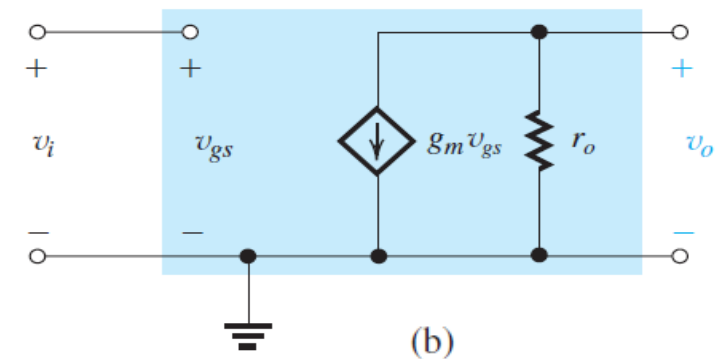
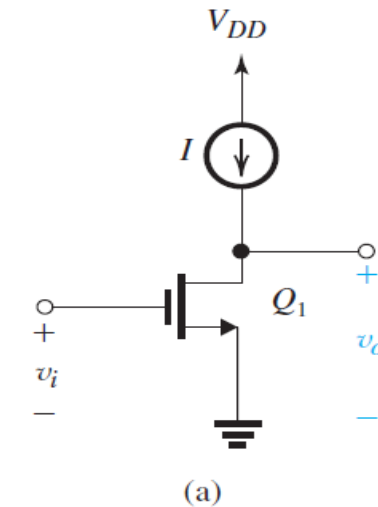
Basic cell/ The CS Amplifiers with Current-Source Loads

- The basic gain cell in an IC amplifier is a common-source (CS) transistor loaded with a constant-current source, as shown in Fig. (a).
- From figure (b), the parameters of CS amplifier are:

$$R_{in} = \infty$$

$$A_{vo} = -g_m r_o$$

$$R_o = r_o$$



a) CS stage amplifier with active load , b) small-signal equivalent circuit of (a)

Basic cell/ The CS Amplifiers with Current-Source Loads

The Intrinsic Gain

- The intrinsic gain ($A_0 = A_{v0}$) of CS stage can given as:

$$A_0 = \frac{V_A}{V_{OV}/2}$$

With

$$g_m = \frac{I_D}{V_{OV}/2}$$

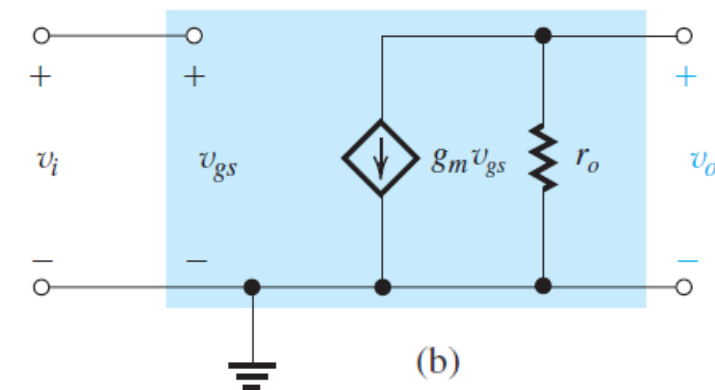
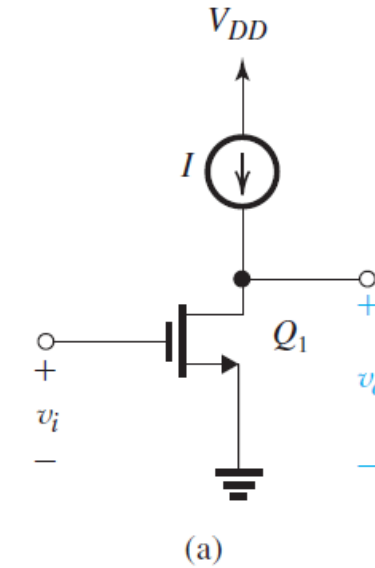
$$g_m = \sqrt{2\mu_n C_{ox}(W/L)}\sqrt{I_D}$$

And

$$r_o = \frac{V_A}{I_D} = \frac{V'_A L}{I_D}$$

Finally we found the intrinsic gain as

$$A_0 = \frac{2V'_A L}{V_{OV}} = \frac{V'_A \sqrt{2(\mu_n C_{ox})(WL)}}{\sqrt{I_D}}$$



a) CS stage amplifier with active load , b) small-signal equivalent circuit of (a)

Basic cell/ The CS Amplifiers with Current-Source Loads

The Intrinsic Gain

- The intrinsic gain is inversely proportional to $\sqrt{I_D}$
- The plot confirms that the gain increases as the bias current is lowered.
- The gain, levels off at very low currents. This is because the MOSFET enters the **subthreshold region** of operation, and the intrinsic gain then becomes constant
- higher gain is obtained at lower values of I_D

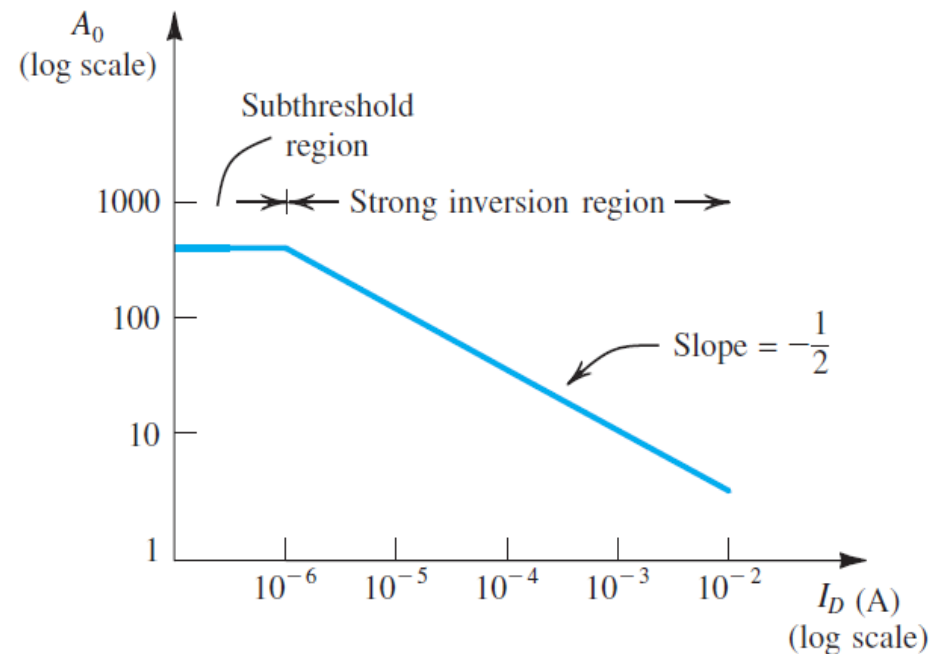
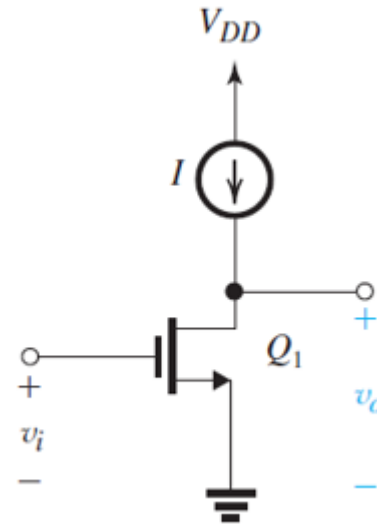


Figure 8.14 The intrinsic gain of the MOSFET versus bias current I_D . Outside the subthreshold region, this is a plot of $A_0 = V_A' \sqrt{2\mu_n C_{ox} WL/I_D}$ for the case: $\mu_n C_{ox} = 20 \mu\text{A/V}^2$, $V_A' = 20 \text{ V}/\mu\text{m}$, $L = 2 \mu\text{m}$, and $W = 20 \mu\text{m}$.

Basic cell/ The CS Amplifiers with Current-Source Loads

Example

Find the values of g_m , R_{in} , R_o , and A_o for a CS amplifier in figure, that is designed using an NMOS transistor with $L = 0.4 \mu\text{m}$ and $W = 4 \mu\text{m}$ and fabricated in a $0.25\text{-}\mu\text{m}$ technology specified to have $\mu_n C_{ox} = 267 \mu\text{A/V}^2$ and $V'_A = 10 \text{ V}/\mu\text{m}$. Assume that the device (transistor) is operating at a drain current of $100 \mu\text{A}$.



Basic cell/ The CS Amplifiers with Current-Source Loads

Example solution

Solution

For simplicity, we shall neglect the Early effect in the MOSFET in determining V_{ov} ; thus,

$$I_D = \frac{1}{2}(\mu_n C_{ox}) \left(\frac{W}{L} \right) V_{ov}^2$$

$$100 = \frac{1}{2} \times 267 \times \left(\frac{4}{0.4} \right) V_{ov}^2$$

resulting in

$$V_{ov} = 0.27 \text{ V}$$

$$g_m = \frac{2I_D}{V_{ov}} = \frac{2 \times 0.1}{0.27} = 0.74 \text{ mA/V}$$

$$R_{in} = \infty$$

$$r_o = \frac{V_A' L}{I_D} = \frac{10 \times 0.4}{0.1} = 40 \text{ k}\Omega$$

$$R_o = r_o = 40 \text{ k}\Omega$$

$$A_0 = g_m r_o = 0.74 \times 40 = 29.6 \text{ V/V}$$

Basic cell/ The CS Amplifiers with Current-Source Loads

Load as a current source

- The current-source load of the CS amplifier is a PMOS biased in saturation region to provide I ;

$$I = \frac{1}{2}(\mu_p C_{ox}) \left(\frac{W}{L}\right)_2 [V_{DD} - V_G - |V_{tp}|]^2$$

- And the current-source load no longer has an infinite resistance; rather, it has a finite output resistance r_{o2} .

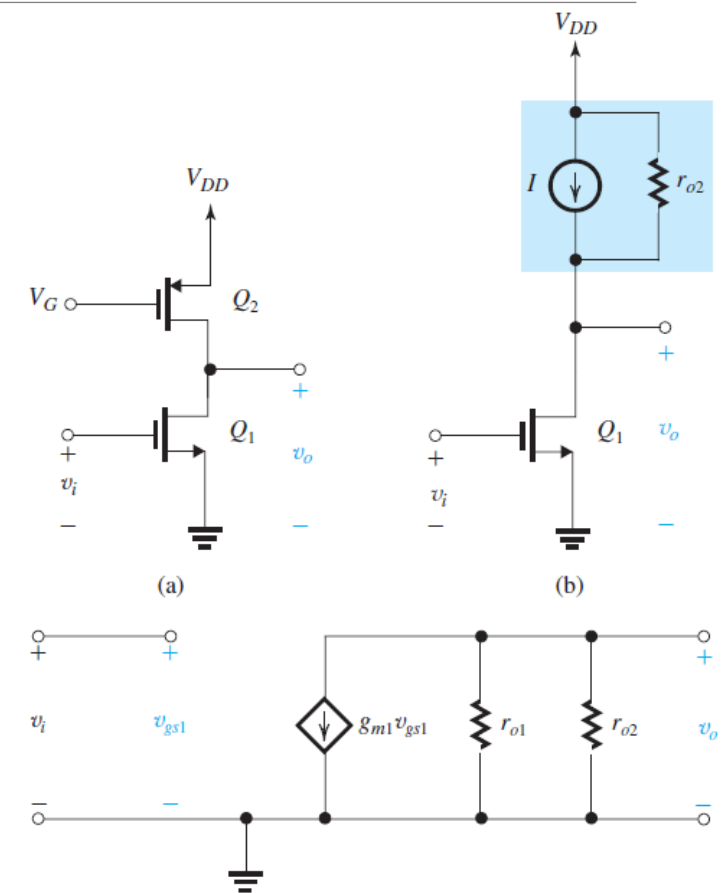
$$r_{o2} = \frac{|V_{A2}|}{I}$$

- from fig (c) we obtain

$$A_v \equiv \frac{v_o}{v_i} = -g_{m1}(r_{o1} \parallel r_{o2})$$

- if Q_2 has an Early voltage equal to that of Q_1 , $r_{o2} = r_{o1}$ and the gain is reduced by half,

$$A_v = -\frac{1}{2}g_m r_o$$

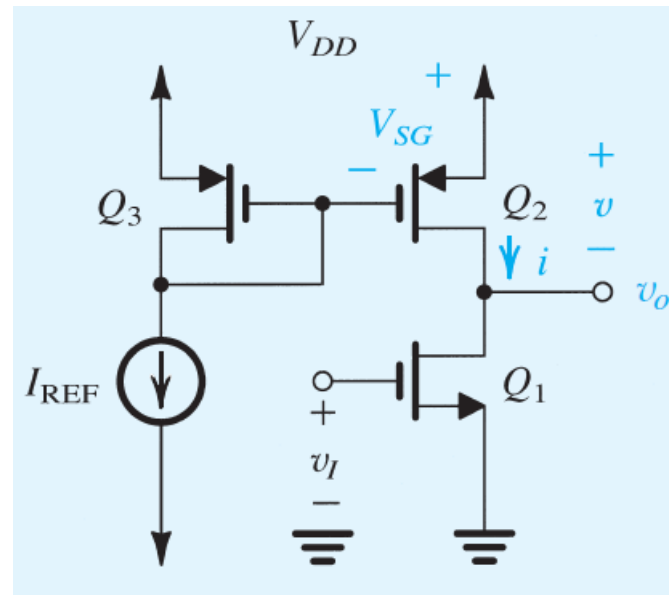


(a) The CS amplifier with the current-source load Q_2 ; (b) the circuit with Q_2 replaced with its large-signal model; (c) small-signal equivalent circuit of the amplifier.

Basic cell/ The CS Amplifiers with Current-Source Loads

Example

Consider the CMOS common-source amplifier in Figure for the case $V_{DD} = 3\text{ V}$, $V_{tn} = V_{tp} = 0.6\text{ V}$, $\mu_n C_{ox} = 200\text{ }\mu\text{A/V}^2$, and $\mu_p C_{ox} = 65\text{ }\mu\text{A/V}^2$. For all transistors, $L = 0.4\text{ }\mu\text{m}$ and $W = 4\text{ }\mu\text{m}$. Also, Early voltage for NMOS and PMOS are respectively $V_{An} = 20\text{ V}$, $V_{Ap} = 10\text{ V}$, and $I_{REF} = 100\text{ }\mu\text{A}$. Find the small-signal voltage gain.



figure

Basic cell/ The CS Amplifiers with Current-Source Loads

Example 1

Solution

$$\begin{aligned}g_{m1} &= \sqrt{2k'_n \left(\frac{W}{L}\right)_1 I_{\text{REF}}} \\&= \sqrt{2 \times 200 \times \frac{4}{0.4} \times 100} = 0.63 \text{ mA/V} \\r_{o1} &= \frac{V_{An}}{I_{D1}} = \frac{20 \text{ V}}{0.1 \text{ mA}} = 200 \text{ k}\Omega \\r_{o2} &= \frac{|V_{Ap}|}{I_{D2}} = \frac{10 \text{ V}}{0.1 \text{ mA}} = 100 \text{ k}\Omega\end{aligned}$$

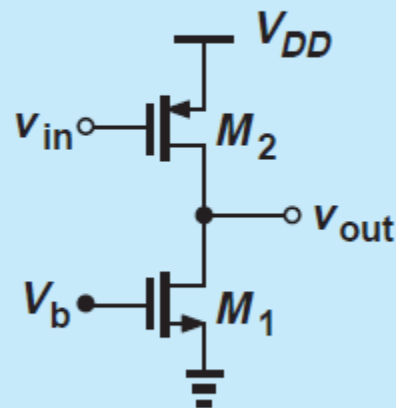
Thus,

$$\begin{aligned}A_v &= -g_{m1}(r_{o1} \parallel r_{o2}) \\&= -0.63(\text{mA/V}) \times (200 \parallel 100)(\text{k}\Omega) = -42 \text{ V/V}\end{aligned}$$

Basic cell/ The CS Amplifiers with Current-Source Loads

Example 2

Figure shows a PMOS CS stage using an NMOS current source load. Compute the voltage gain of the circuit.



CS stage using an NMOS device as current source.

Solution Transistor M_2 generates a small-signal current equal to $g_{m2}v_{in}$, which then flows through $r_{O1}||r_{O2}$, producing $v_{out} = -g_{m2}v_{in}(r_{O1}||r_{O2})$. Thus,

$$A_v = -g_{m2}(r_{O1}||r_{O2}).$$

Basic cell/ The CS Amplifiers with source degeneration

Large Signal model:

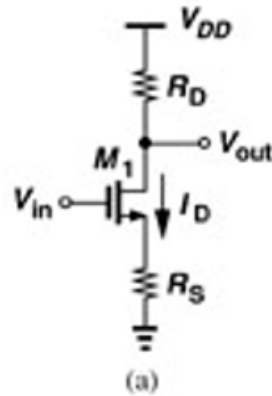
$$G_m = \frac{\partial I_D}{\partial V_{in}} = \frac{\partial I_D}{\partial V_{GS}} \frac{\partial V_{GS}}{\partial V_{in}}$$

$$V_{GS} = V_{in} - I_D R_S$$

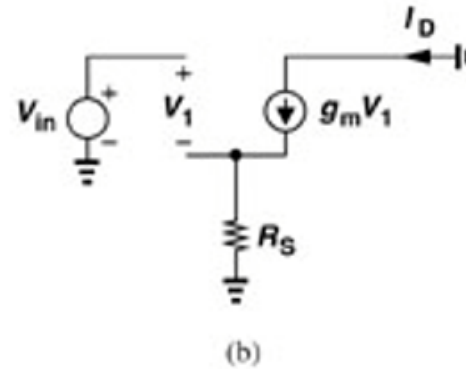
$$\frac{\partial V_{GS}}{\partial V_{in}} = 1 - \frac{\partial I_D}{\partial V_{in}} R_S$$

$$G_m = \frac{\partial I_D}{\partial V_{GS}} \left(1 - R_S \frac{\partial I_D}{\partial V_{in}} \right)$$

$$G_m = g_m (1 - R_S G_m)$$



Small Signal model:



$$G_m = \frac{I_D}{V_{in}} = \frac{g_m V_1}{V_1 + g_m V_1 R_S}$$

$$G_m = \frac{g_m}{1 + g_m R_S} = \frac{g_m}{1 + g_m R_S} = \frac{1}{1/g_m + R_S}$$

for $R_S \gg 1/g_m \Rightarrow G_m \approx 1/R_S$

$$G_m = \frac{g_m}{1 + g_m R_S}$$

$$A_v = -G_m R_D$$

$$A_v = -\frac{g_m R_D}{1 + g_m R_S}$$

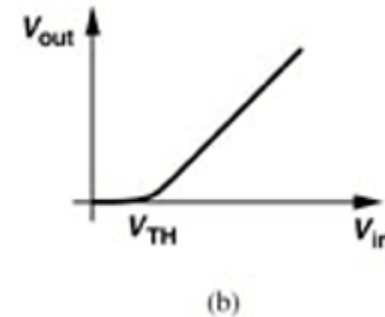
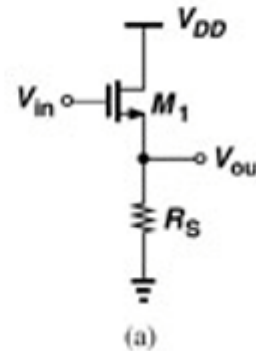
Basic cell/ The Source follower Amplifiers (CD: Common Drain)

Large Signal Behavior

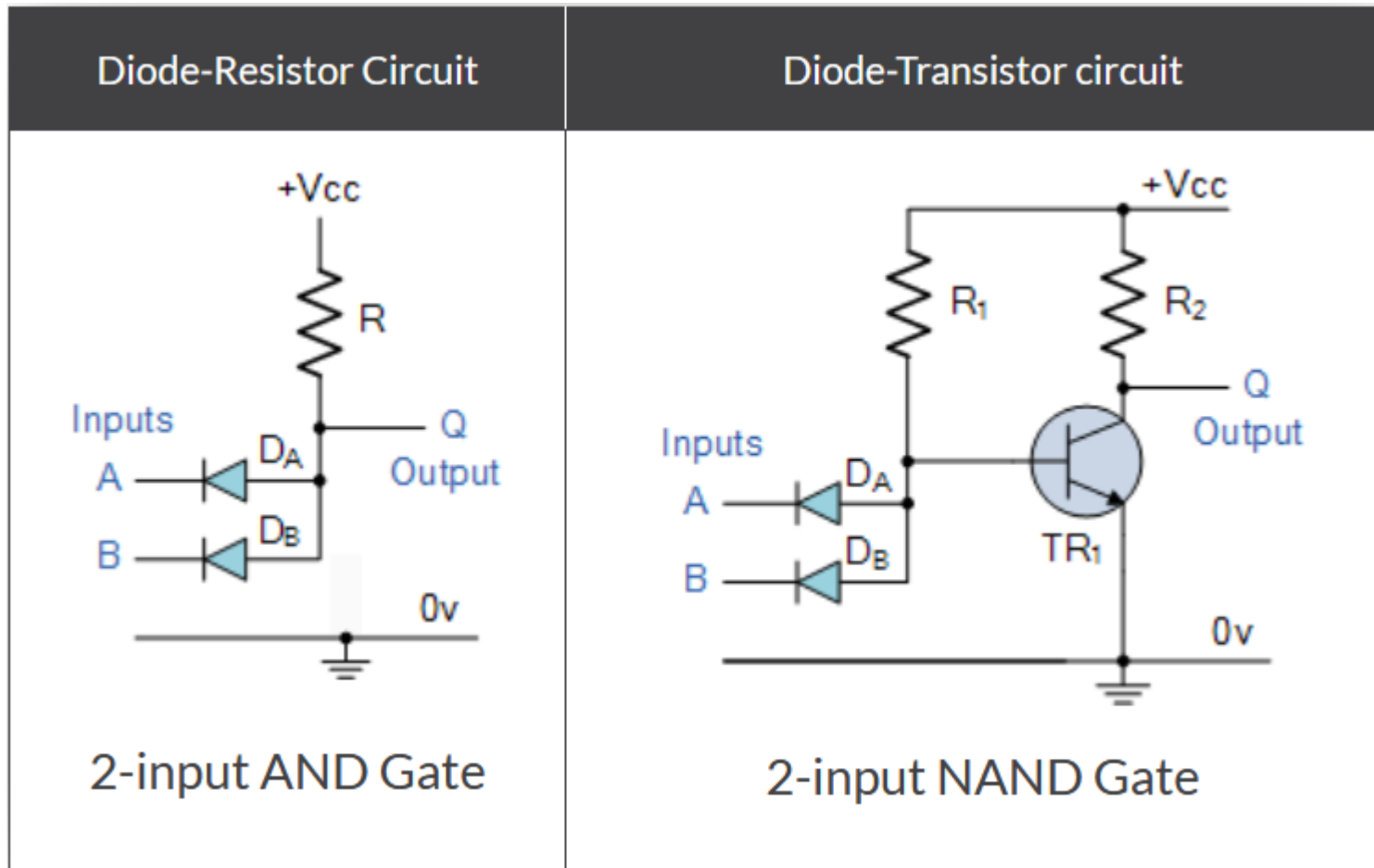
M1 turns on in saturation:

$$V_{out} = I_D R_S$$

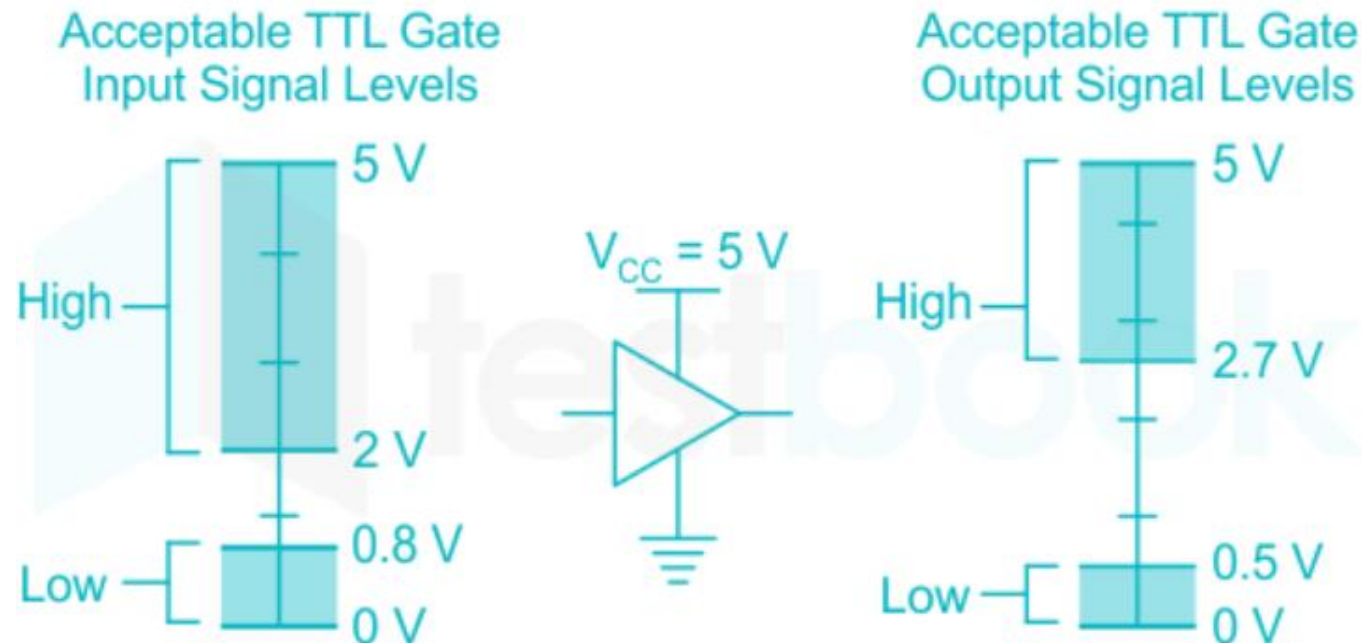
$$V_{out} = \frac{\mu_n C_{ox}}{2} \frac{W}{L} (V_{in} - V_{out} - V_{TH})^2 R_S$$



Before CMOS Logic Circuits



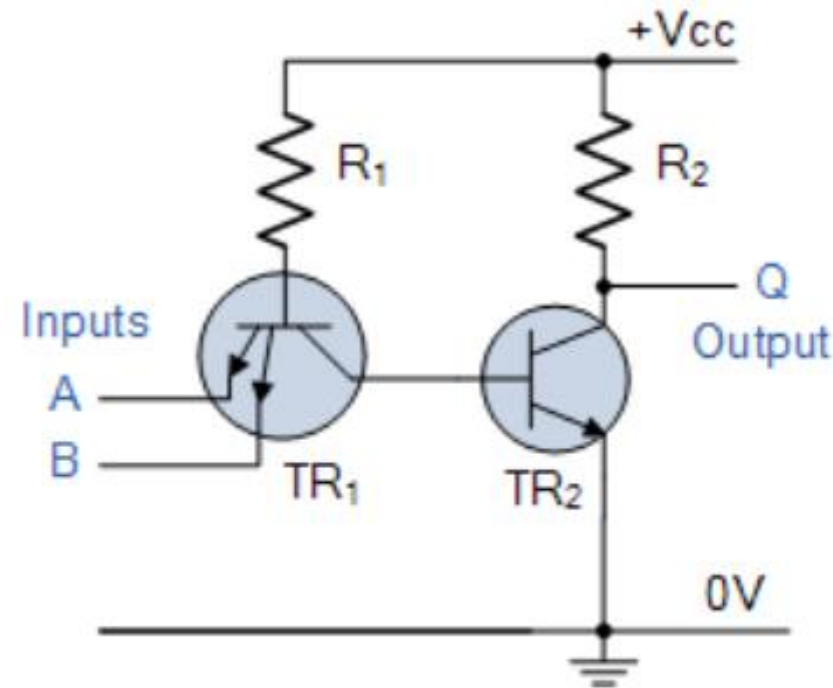
Before CMOS Logic Circuits



- TTL outputs are typically restricted to narrower limits of between 0.0 V and 0.4 V for a "low".
- TTL outputs are typically restricted to narrower limits of between 2.4 V and 5 V for a "high", providing at least 0.4 V of noise immunity.

Before CMOS Logic Circuits

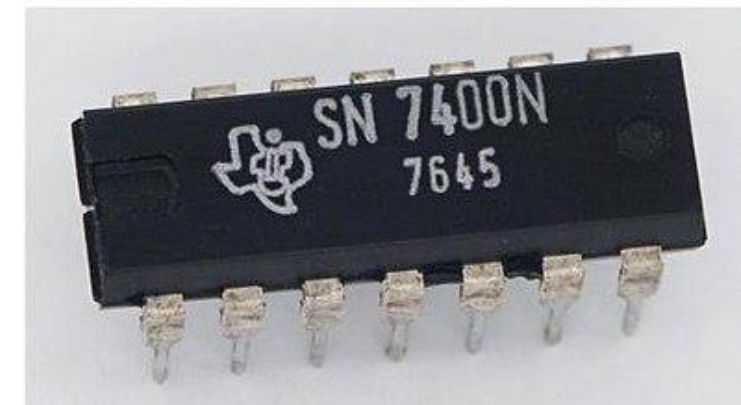
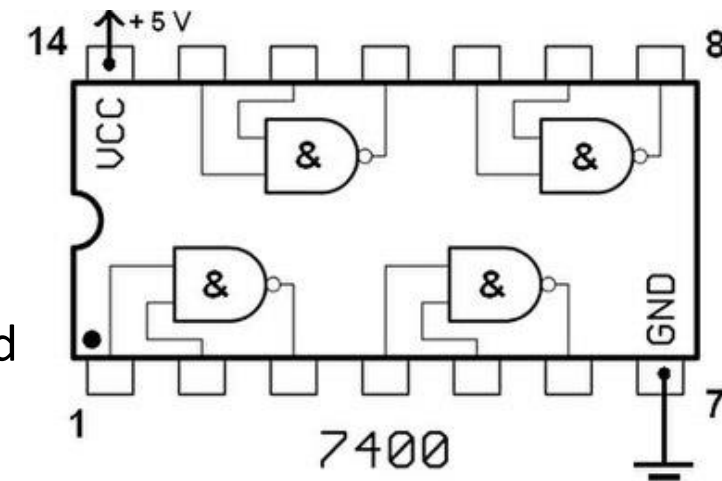
TTL was *transistor-coupled transistor logic* (TCTL).



2-input NAND Gate

Before CMOS Logic Circuits

TTL was invented in 1961 by [James L. Buie](#) of [TRW](#), which declared it "particularly suited to the newly developing integrated circuit design technology." The original name for TTL was *transistor-coupled transistor logic* (TCTL).^[2] The first commercial integrated-circuit TTL devices were manufactured by Sylvania in 1963, called the Sylvania Universal High-Level Logic family (SUHL).^[3] The Sylvania parts were used in the controls of the [Phoenix missile](#).^[3] TTL became popular with electronic systems designers after [Texas Instruments](#) introduced the 5400 series of ICs, with military temperature range, in 1964 and the later [7400 series](#), specified over a narrower range and with inexpensive plastic packages, in 1966.^[4]

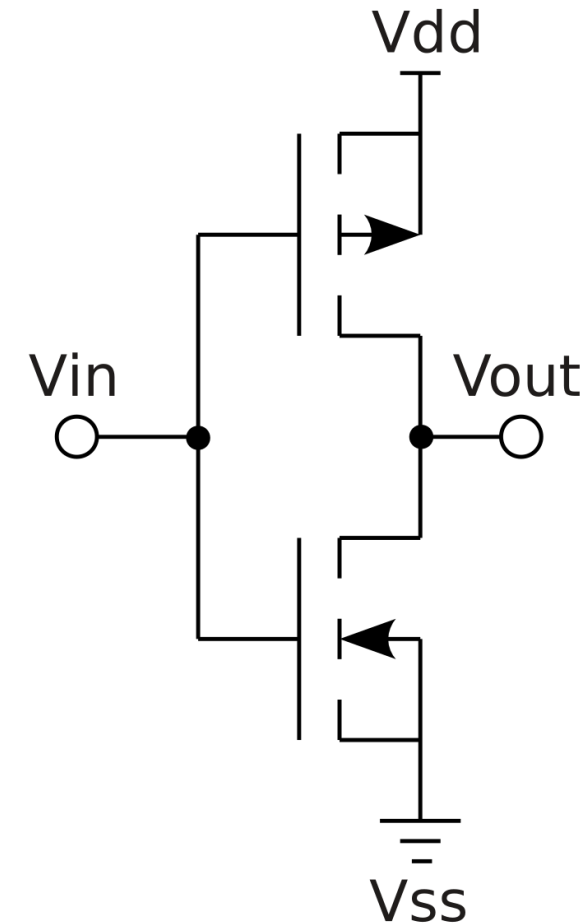


Wikipedia

CMOS Logic Circuits

[Mohamed Atalla](#) and [Dawon Kahng](#) proposed a silicon MOS transistor in 1959^[17] and successfully demonstrated a working MOS device with their Bell Labs team in 1960.^{[18][19]} Their team included E. E. LaBate and E. I. Povolonis who fabricated the device; M. O. Thurston, L. A. D'Asaro, and J. R. Ligenza who developed the diffusion processes, and H. K. Gummel and R. Lindner who characterized the device.^{[20][21]} There were originally two types of MOSFET logic, [PMOS](#) ([p-type](#) MOS) and [NMOS](#) ([n-type](#) MOS).^[22] Both types were developed by Frosch and Derrick in 1957 at Bell Labs.^[23]

CMOS was commercialised by [RCA](#) in the late 1960s. RCA adopted CMOS for the design of [integrated circuits](#) (ICs), developing CMOS circuits for an [Air Force](#) computer in 1965 and then a 288-bit CMOS [SRAM](#) memory chip in 1968.^[25] RCA also used CMOS for its [4000-series integrated circuits](#) in 1968, starting with a 20 [µm semiconductor manufacturing process](#) before gradually scaling to a [10 µm process](#) over the next several years.^[28]



CMOS Logic Circuits

- MOS-based technology has become the default standard for high-density logic designs for several reasons.
- The most obvious is that MOSFETs can be made with side dimensions of $<5\text{nm}$, allowing for complex logic functions to be constructed in small areas.
- The section is an investigation of the basics of designing and characterizing logic gates in an MOS technology.

MOS Logic Circuits

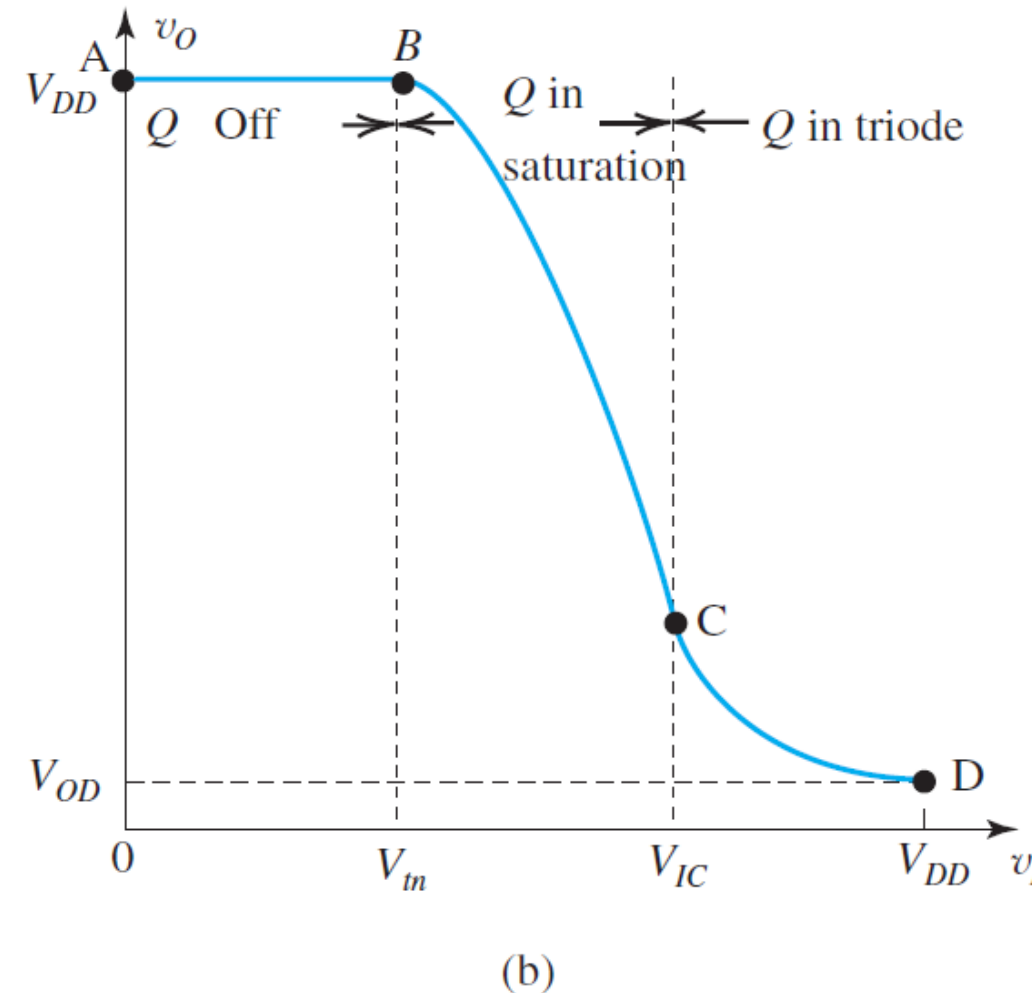
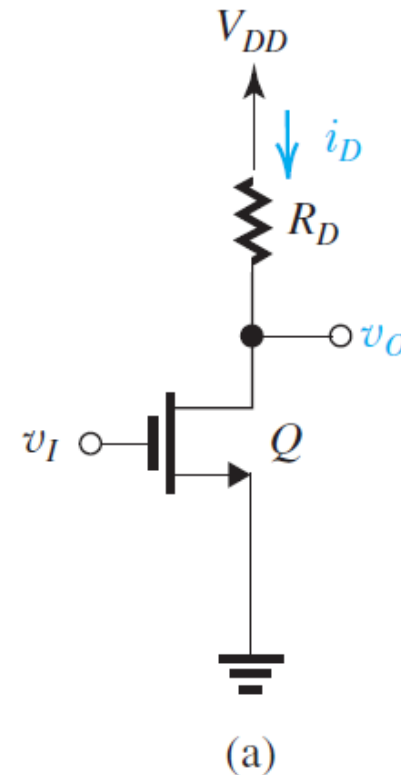
The simple resistively loaded MOS amplifier can be used as a logic inverter

when operated in cutoff $v_I < V_{tn}$

and in triode

$v_I > V_{IC}$

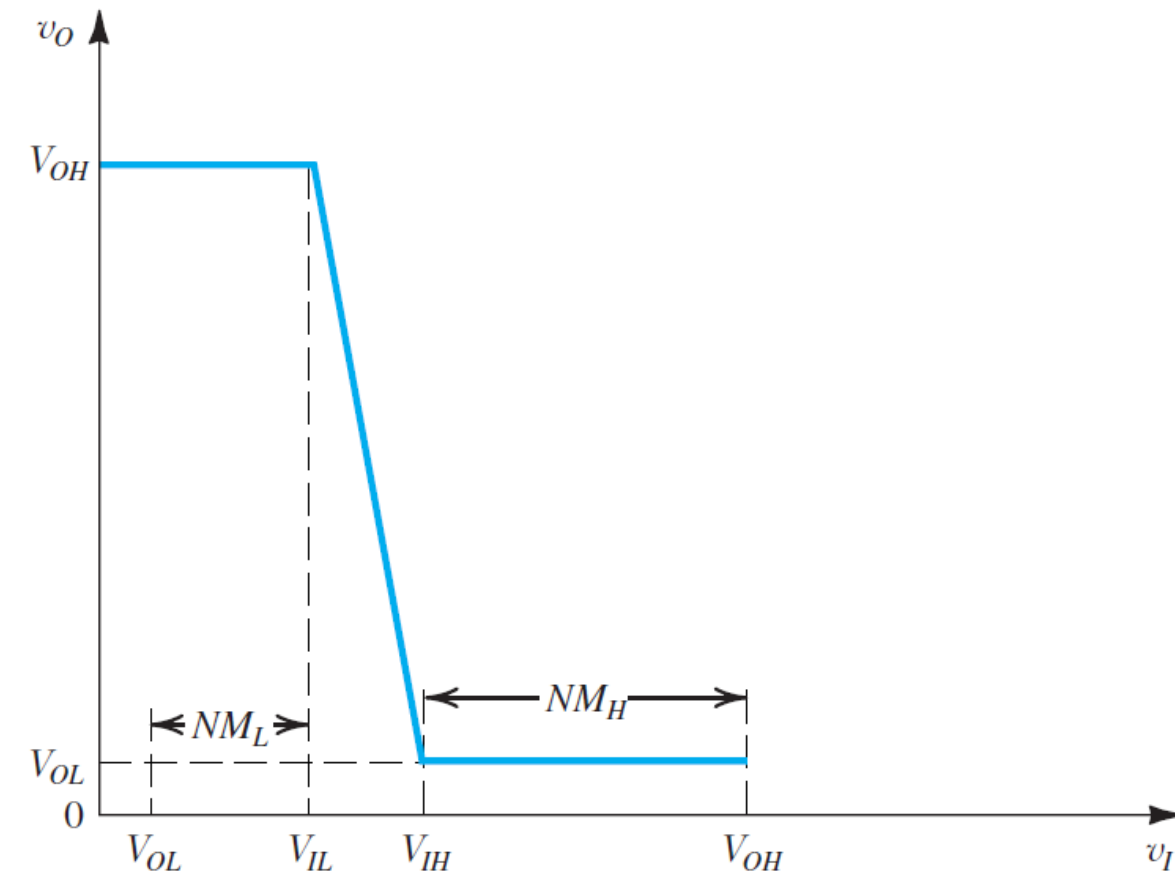
The output high level is V_{DD} and the low level is V_{OD}



MOS Logic Circuits

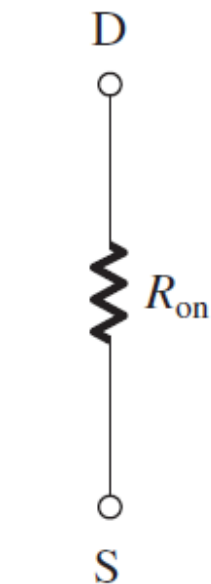
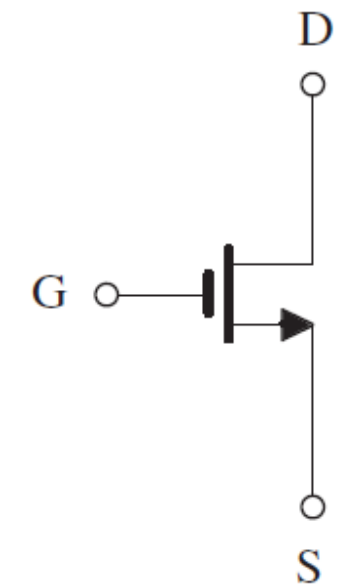
- The VTC is approximated by three straight-line segments.
- The four parameters of the VTC (V_{OH} , V_{OL} , V_{IL} , and V_{IH}) and their use in determining the noise margins (NM_H and NM_L).

V_{OL} : Output low level
 V_{OH} : Output high level
 V_{IL} : Maximum value of input interpreted by the inverter as a logic 0
 V_{IH} : Minimum value of input interpreted by the inverter as a logic 1
 NM_L : Noise margin for low input = $V_{IL} - V_{OL}$
 NM_H : Noise margin for high input = $V_{OH} - V_{IH}$



Voltage-transfer characteristic of an inverter.

MOSFET Models for Digital Circuits



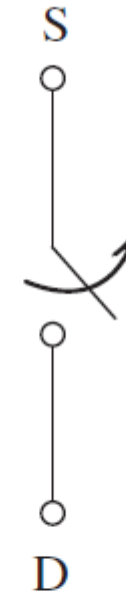
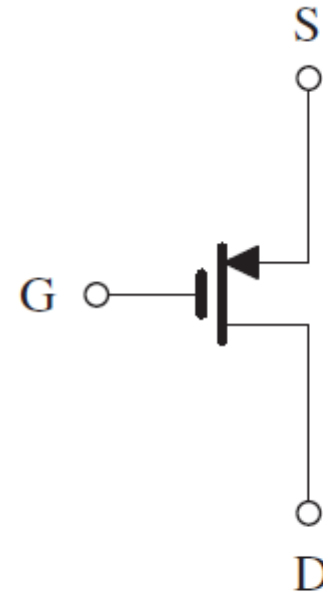
$$V_G = V_{DD}$$

$$(G = 1)$$



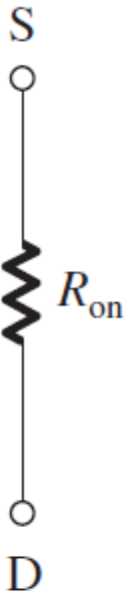
$$V_G = 0$$

$$(G = 0)$$



$$V_G = V_{DD}$$

$$(G = 1)$$

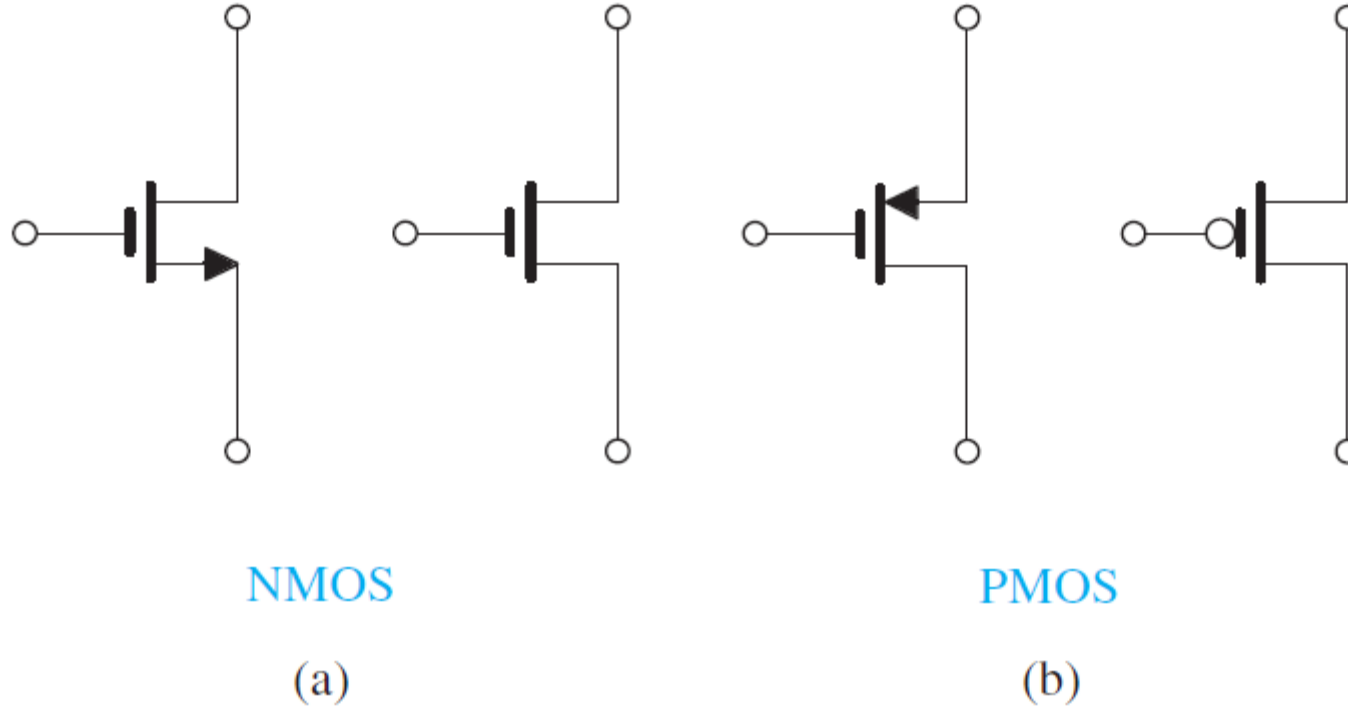


$$V_G = 0$$

$$(G = 0)$$

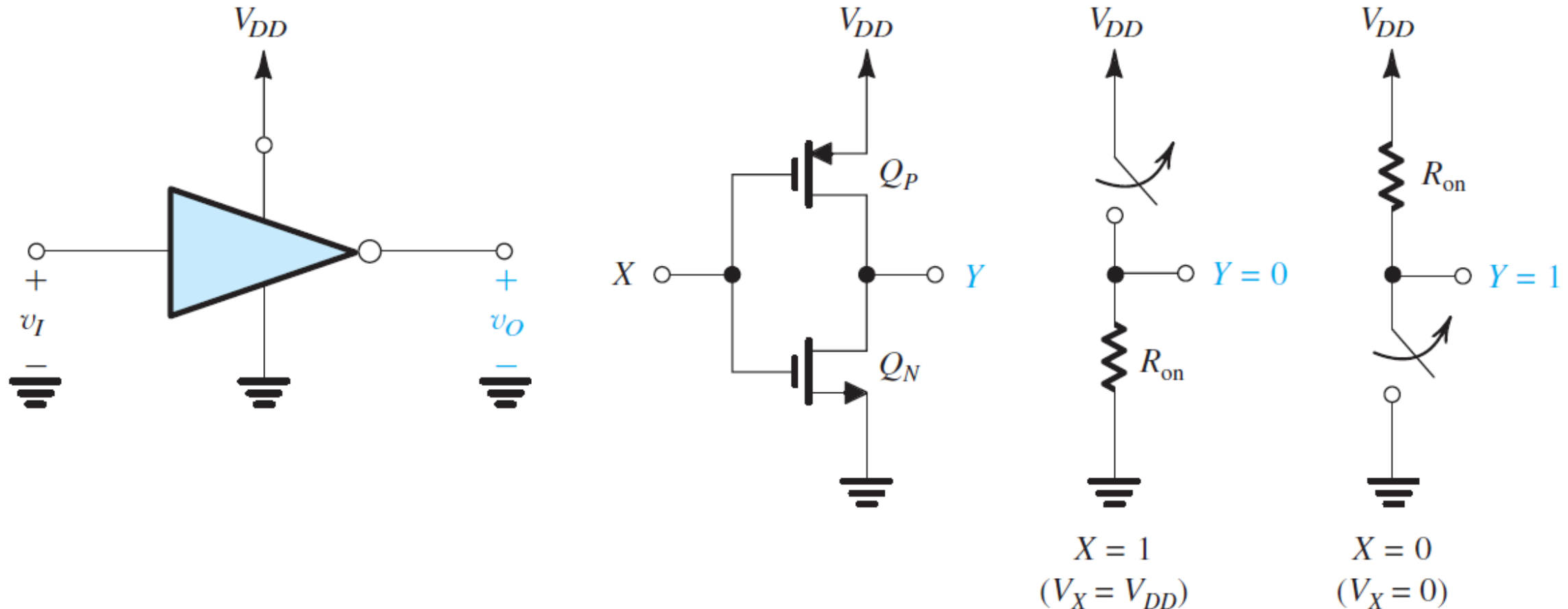
Operation of the NMOS and PMOS transistor as an on/off switch.

MOSFET Models for Digital Circuits



Usual NMOS and PMOS transistor as an on/off switch.

The CMOS Inverter



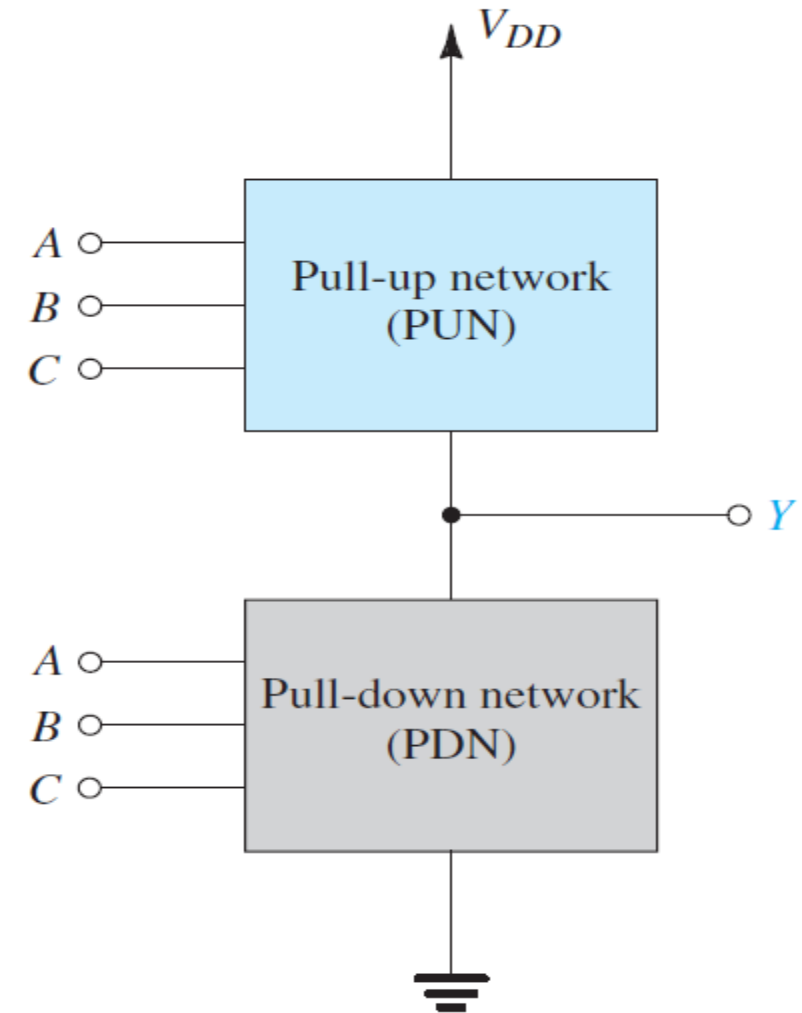
logic inverter; (b) its CMOS realization; (c) operation when the input is a logic 1; (d) operation when the input is a logic 0.

General Structure of CMOS Logic

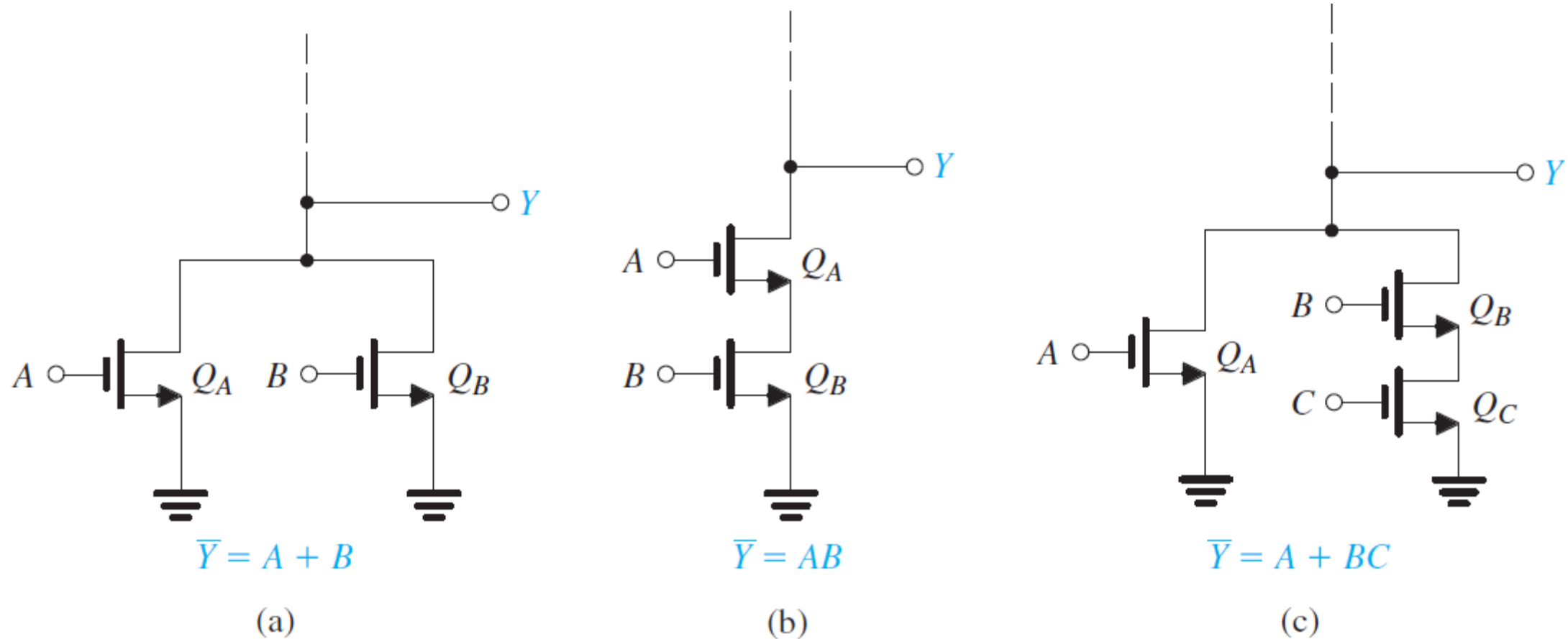
Representation of a three-input CMOS logic gate.

The PUN comprises PMOS transistors, and the PDN

comprises NMOS transistors.

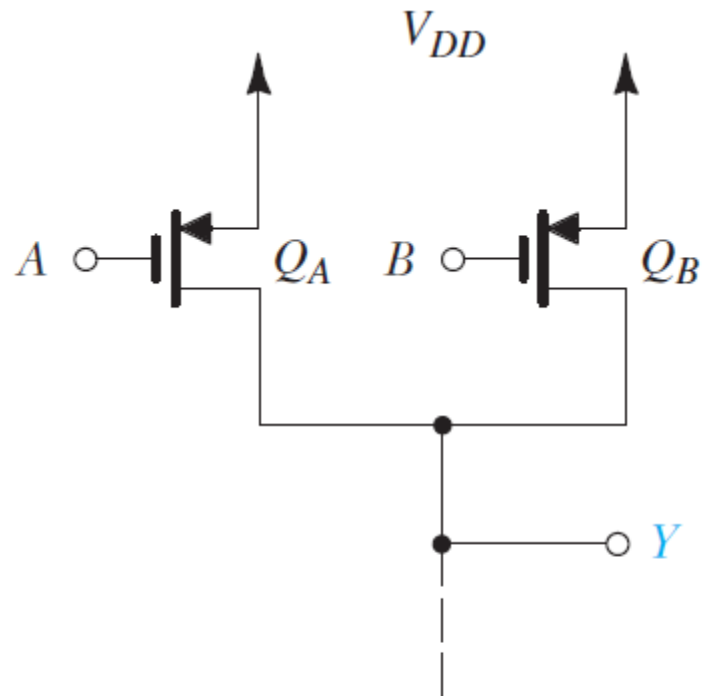


General Structure of CMOS Logic



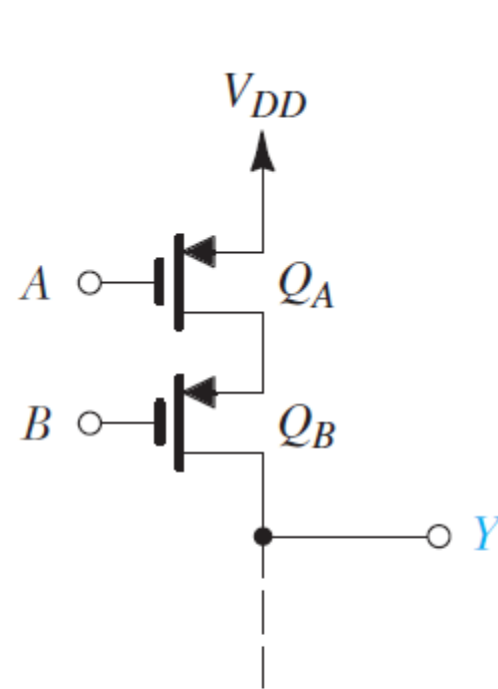
Examples of pull-down networks.

General Structure of CMOS Logic



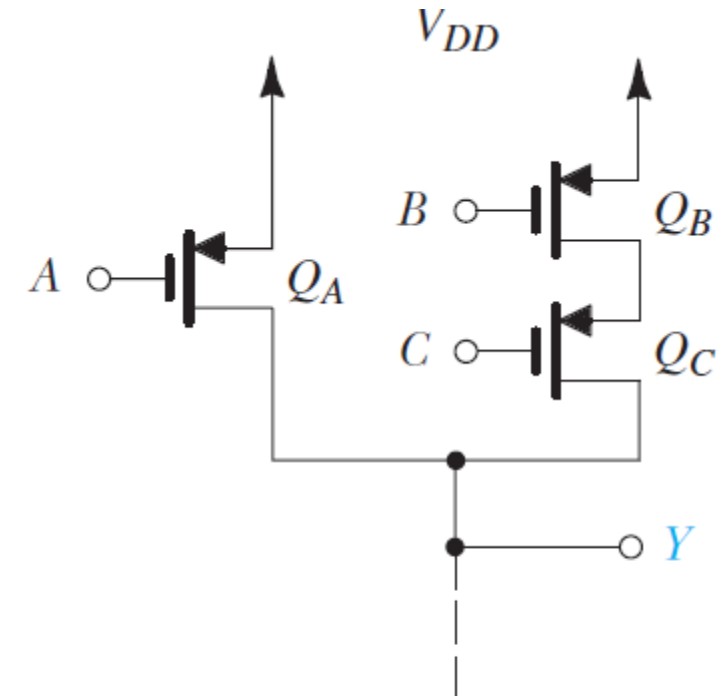
$$Y = \overline{A} + \overline{B}$$

(a)



$$Y = \overline{A} \overline{B}$$

(b)



$$Y = \overline{A} + \overline{B} \overline{C}$$

(c)

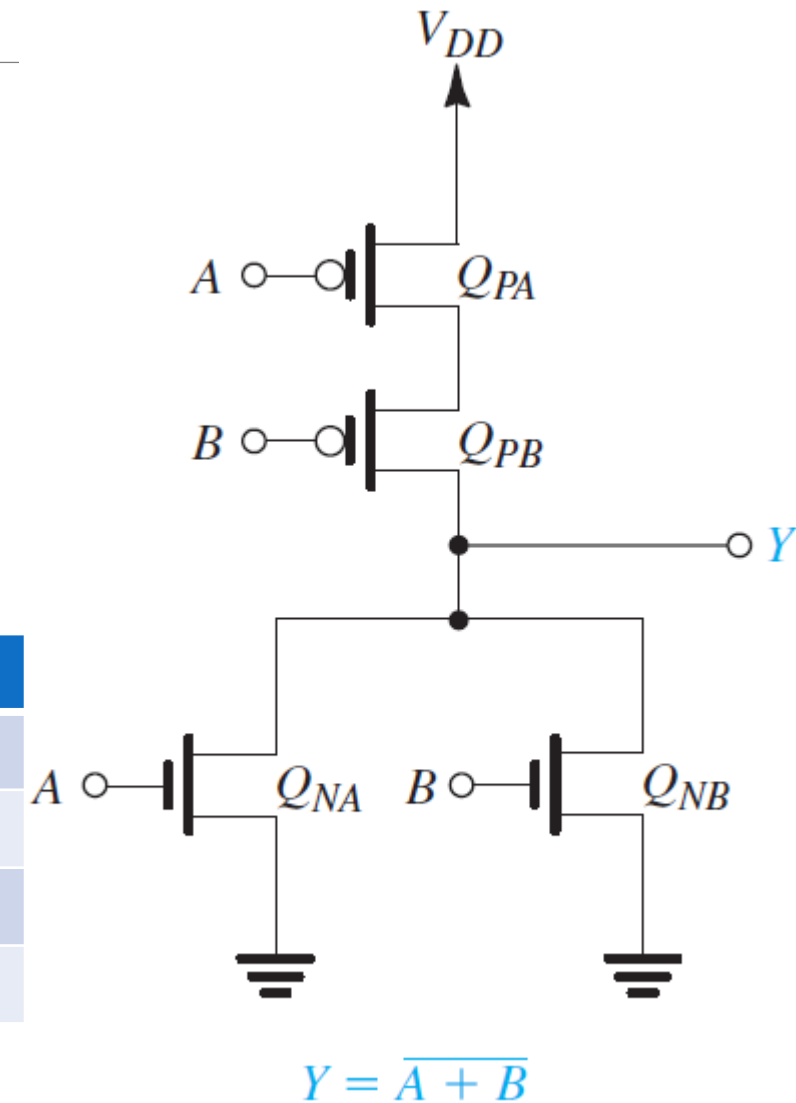
Examples of pull-up networks.

Two-Input NOR Gate

The CMOS gate that realizes the two-input NOR function

$$Y = \overline{A + B} = \overline{A} \overline{B}$$

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0



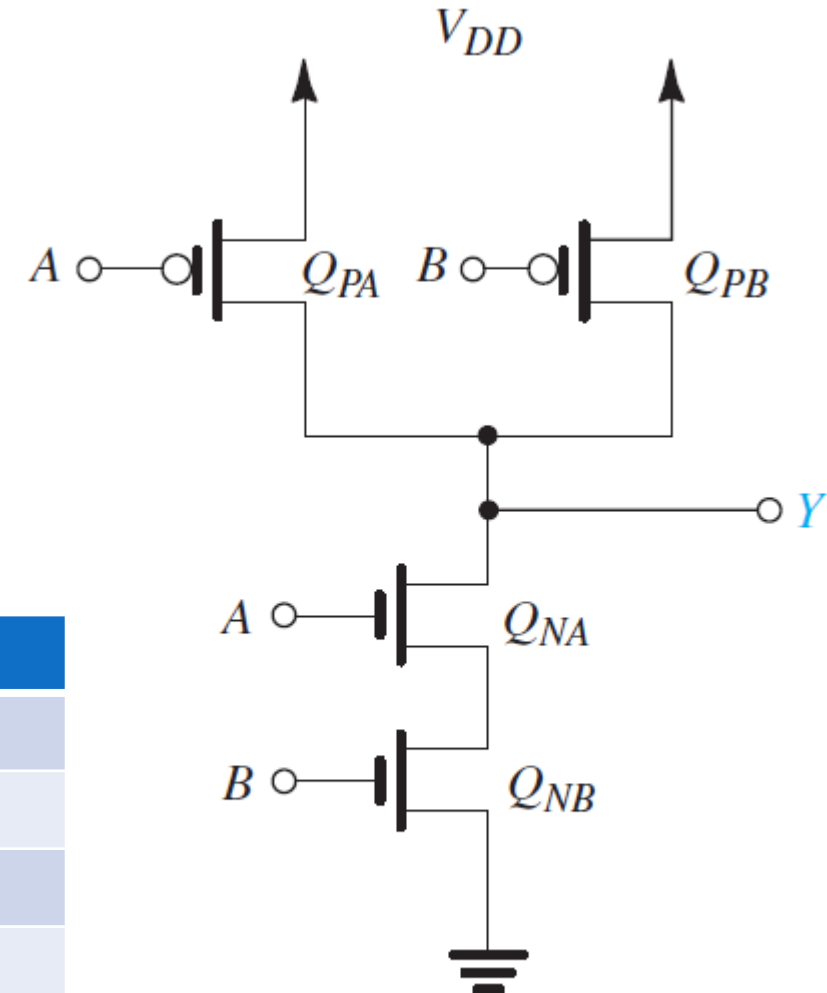
Two-Input NAND Gate

The CMOS gate that realizes the two-input NAND function

$$Y = \overline{AB} = \overline{A} + \overline{B}$$

Truth table

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0



$$Y = \overline{AB}$$

A Complex Gate

Consider next the more complex logic function

$$Y = \overline{A(B + CD)}$$

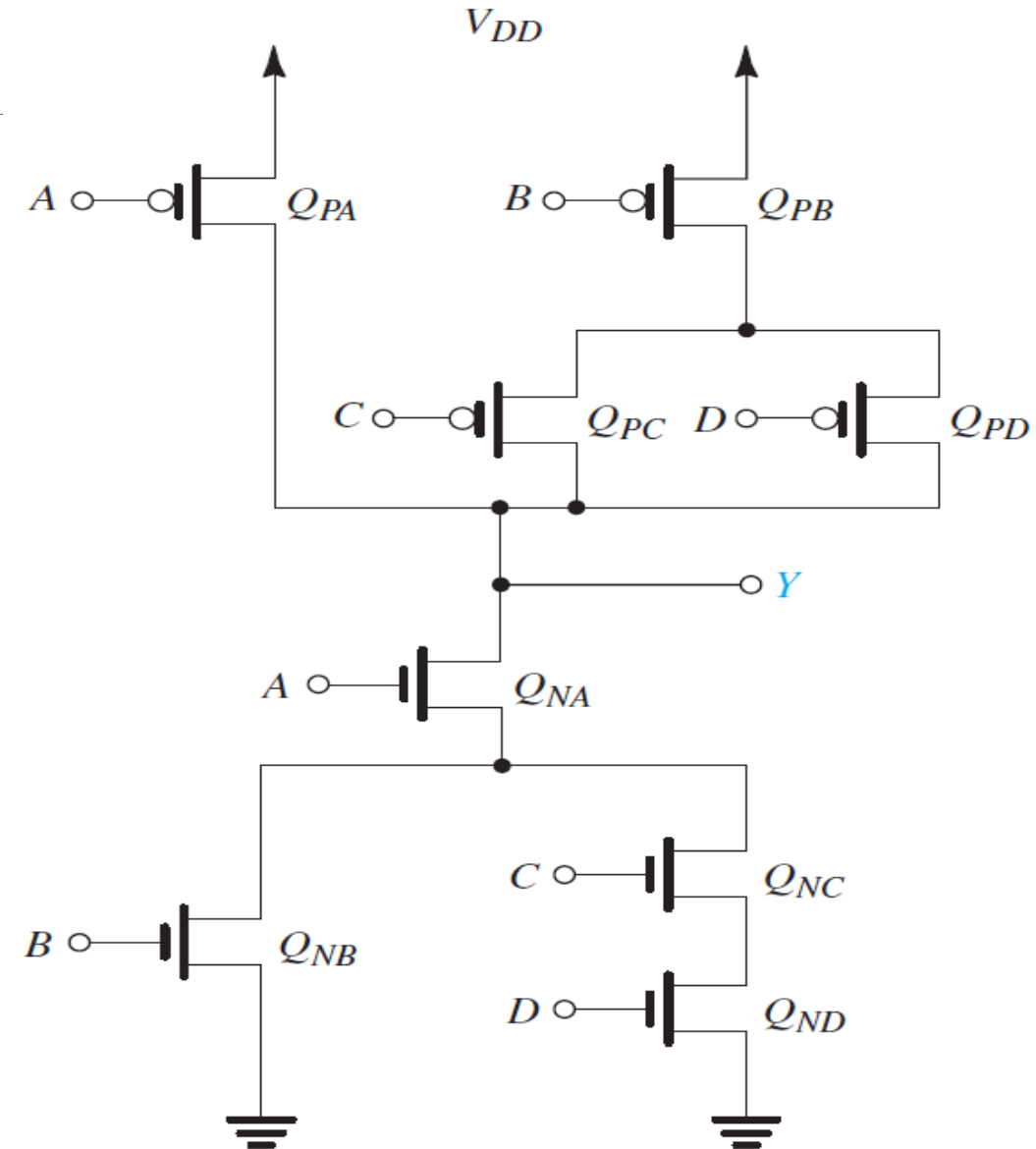
Since $\overline{Y} = A(B + CD)$, we see that Y should be low for A high and simultaneously either B high or C and D both high, from which the PDN is directly obtained. To obtain the PUN, we need to express Y in terms of the complemented variables. We do this through repeated application of DeMorgan's law, as follows:

$$\begin{aligned} Y &= \overline{A(B + CD)} \\ &= \overline{A} + \overline{B + CD} \\ &= \overline{A} + \overline{B} \overline{CD} \\ &= \overline{A} + \overline{B} (\overline{C} + \overline{D}) \end{aligned}$$

Thus, Y is high for A low or B low and either C or D low.

A Complex Gate

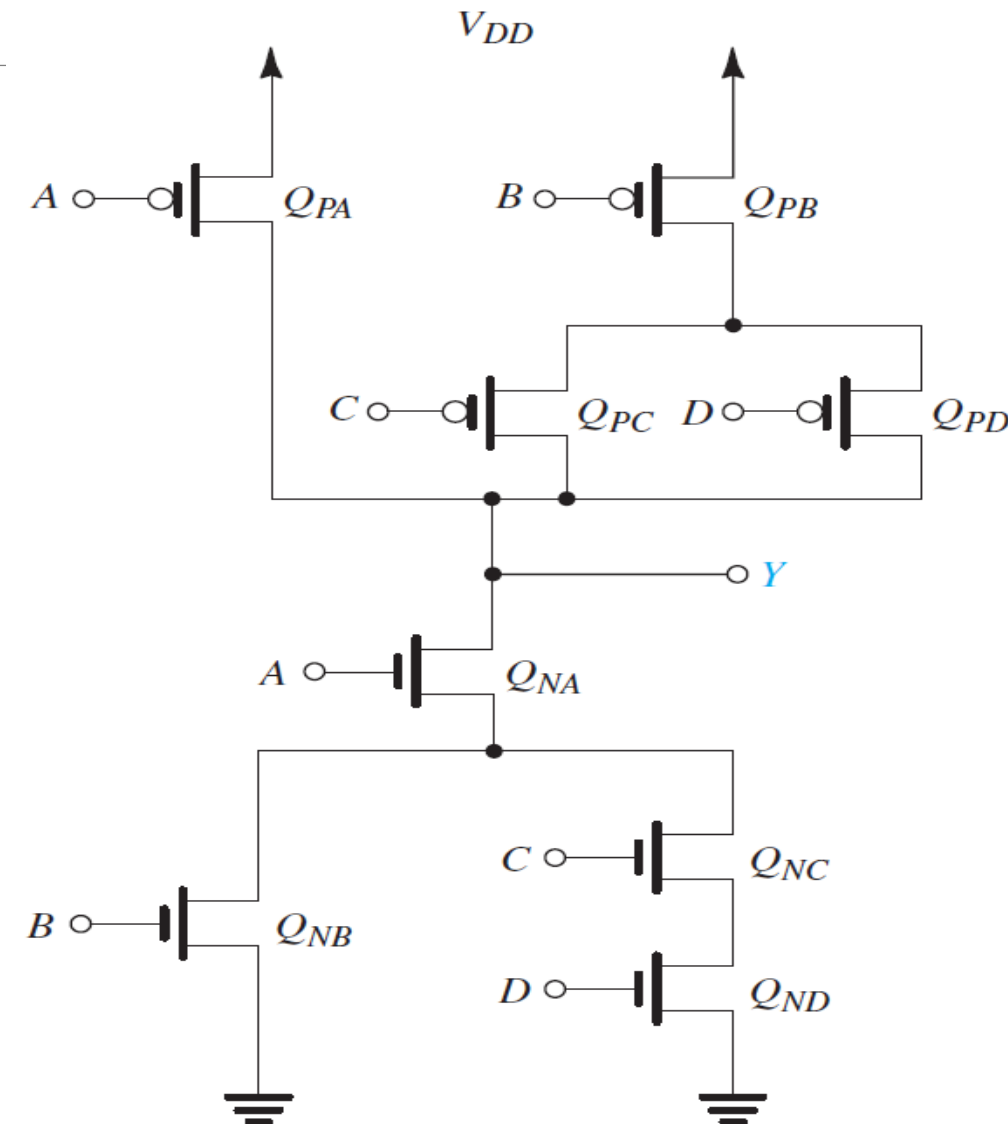
$$\begin{aligned}Y &= \overline{A(B + CD)} \\&= \overline{A} + \overline{B + CD} \\&= \overline{A} + \overline{B} \overline{CD} \\&= \overline{A} + \overline{B}(\overline{C} + \overline{D})\end{aligned}$$



Obtaining the PUN from the PDN and Vice Versa

From the CMOS gate circuits considered thus far (e.g., Fig. at right), we observe that the PDN and the PUN are dual networks: Where a series branch exists in one, a parallel branch exists in the other. Thus, we can obtain one from the other, a process that can be simpler than having to synthesize each separately from the Boolean expression of the function.

$$Y = \overline{A(B + CD)}$$



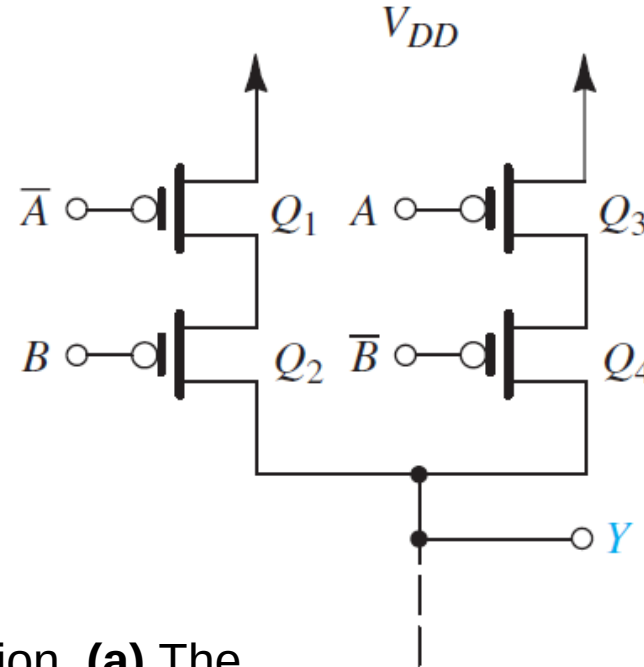
Exclusive-OR Function

An important function that often arises in logic design is the exclusive-OR (XOR) function,

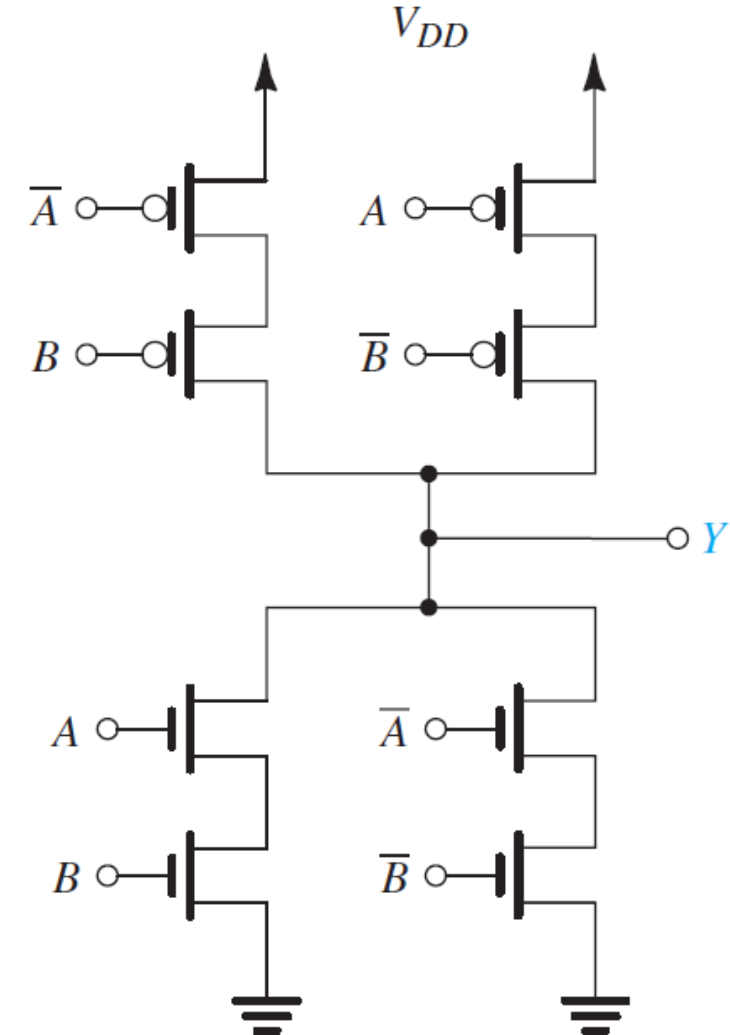
$$Y = A\bar{B} + \bar{A}B$$

$$\bar{Y} = AB + \bar{A}\bar{B}$$

Realization of the exclusive-OR (XOR) function. **(a)** The PUN synthesized directly from the expression in Eq. (Y). **(b)** The complete XOR realization utilizing the PUN in **(a)** and a PDN that is synthesized directly from the expression in Eq. ().



(a)



(b)